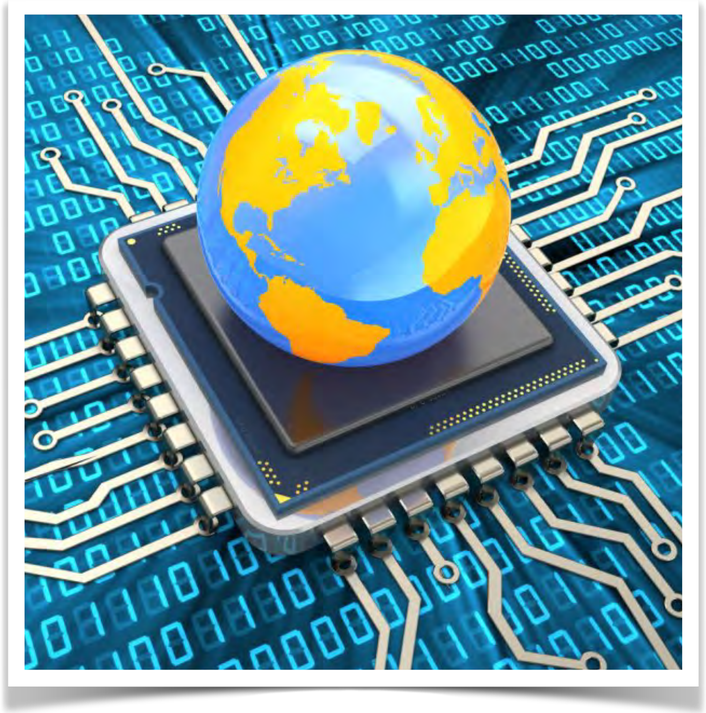
overview - week 1



computer architecture

overview - week 2



system
software

overview - week 3



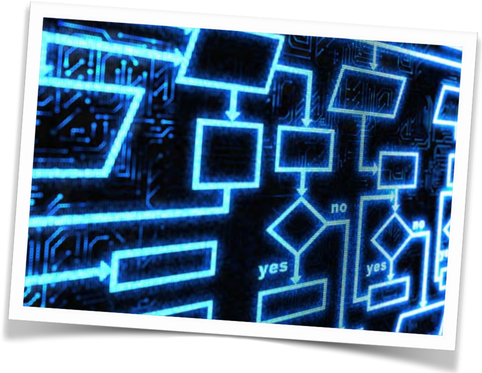
programming basics

python

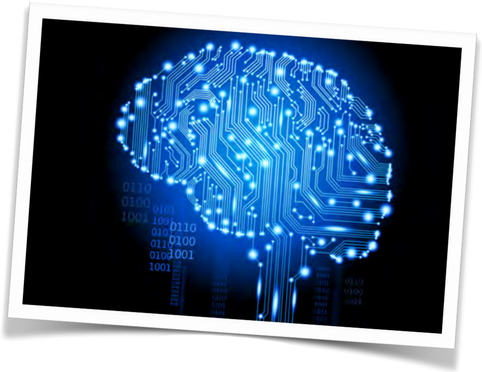
java

overview - week 4

functions



memory



&

exceptions



overview - week 5



collections,
iteration &
recursion

overview - week 6



object-oriented programming

part 1

overview - week 7



object-oriented programming

part 2

overview - week 8

mid-term test



overview - week 9

mid-term test
correction



overview - week 9

mid-term test

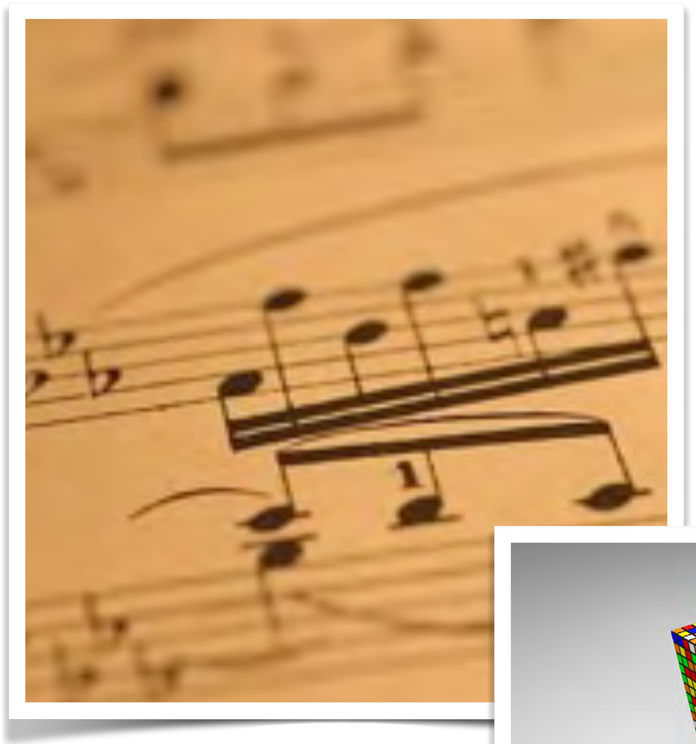
correction

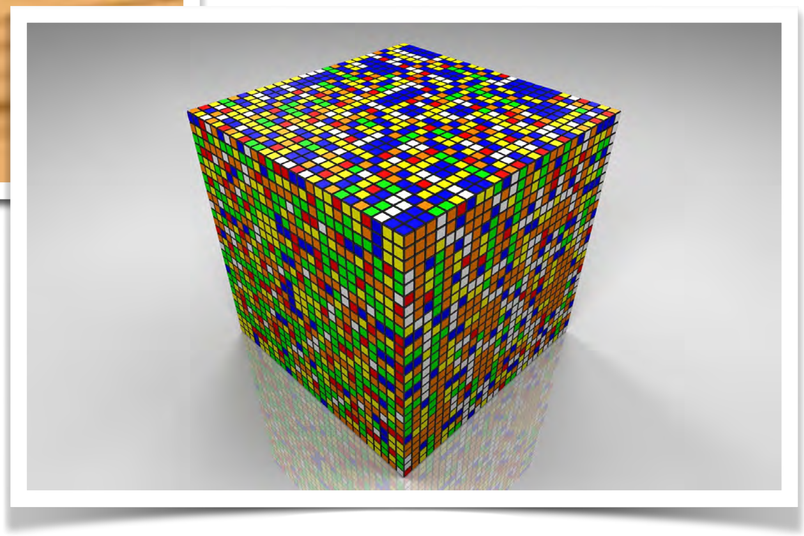
& discrete math





overview - week 10





algorithms & computational complexity

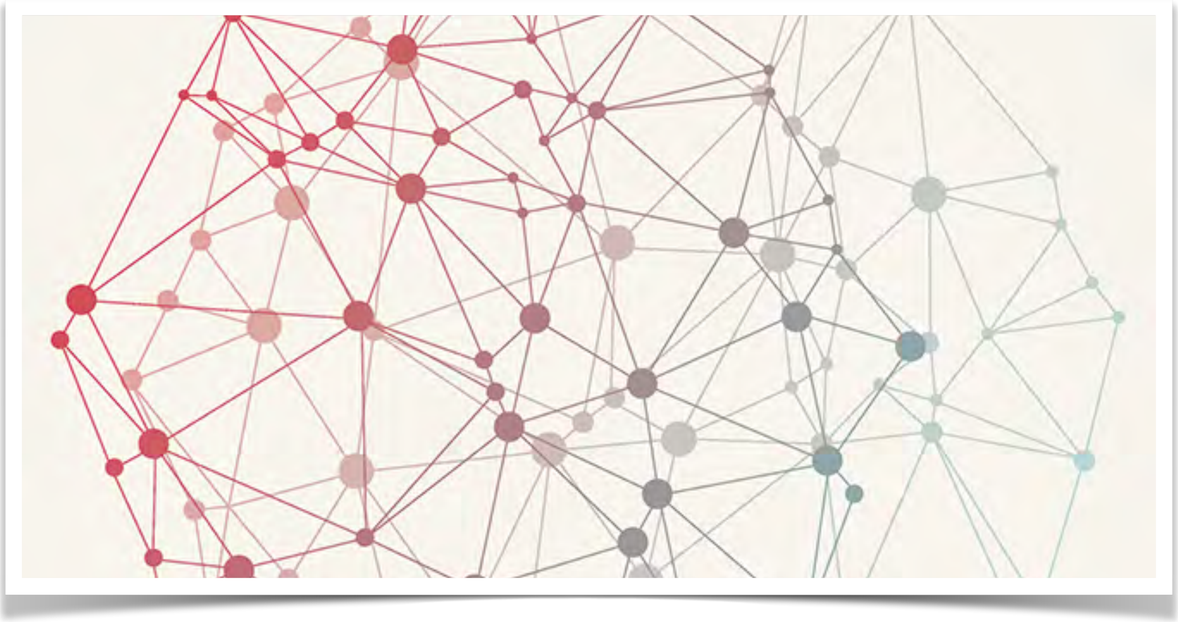
overview - week 11



search

algorithms

overview - week 12



graph

algorithms

overview - week 13



spatial tree

algorithms

overview - week 14



probabilistic algorithms

calendar

TUESDAY	8.00–10.00		Group A :14.15–16.00 + Group B :16.15–18.00
Sep 15	course overview	computer architecture	exercises on computer architecture
Sep 22	system software		exercises on system software
Sep 29	programming basics		exercises on programming basics
Oct 06	functions, memory and exceptions		exercises on functions, memory and exceptions
Oct 13	collections, iteration and recursion		exercises on collections, iteration and recursion
Oct 20	object-oriented programming – part 1		exercises on object-oriented programming – part 1
Oct 27	object-oriented programming – part 2		**consolidation exercises**
Nov 03	**mid-term test**		exercises on object-oriented programming – part 2
Nov 10	**mid-term test correction**	discrete math	exercises on discrete math
Nov 17	algorithms and their complexity		exercises on algorithms and their complexity
Nov 24	search algorithms		exercises on search algorithms
Dec 01	graphs algorithms		exercises on graph algorithms
Dec 08	spatial tree algorithms		exercises on spatial tree algorithms
Dec 15	probabilistic algorithms		exercises on probabilistic algorithms

so many whys...

- why two programming languages?
- why pseudo-code in addition?
- why no video recordings?
- why English slides?
- why so many slides?

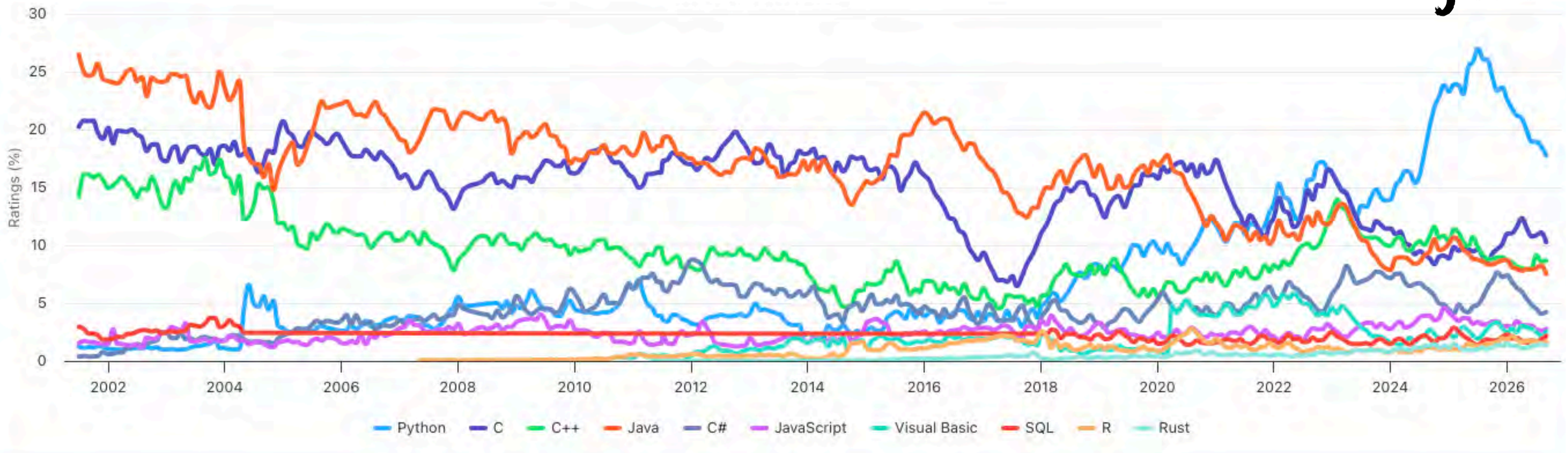


programming languages



TIOBE Programming Community Index Definition

https://www.tiobe.com/tiobe-index



The calculation consists in counting hits for the search query

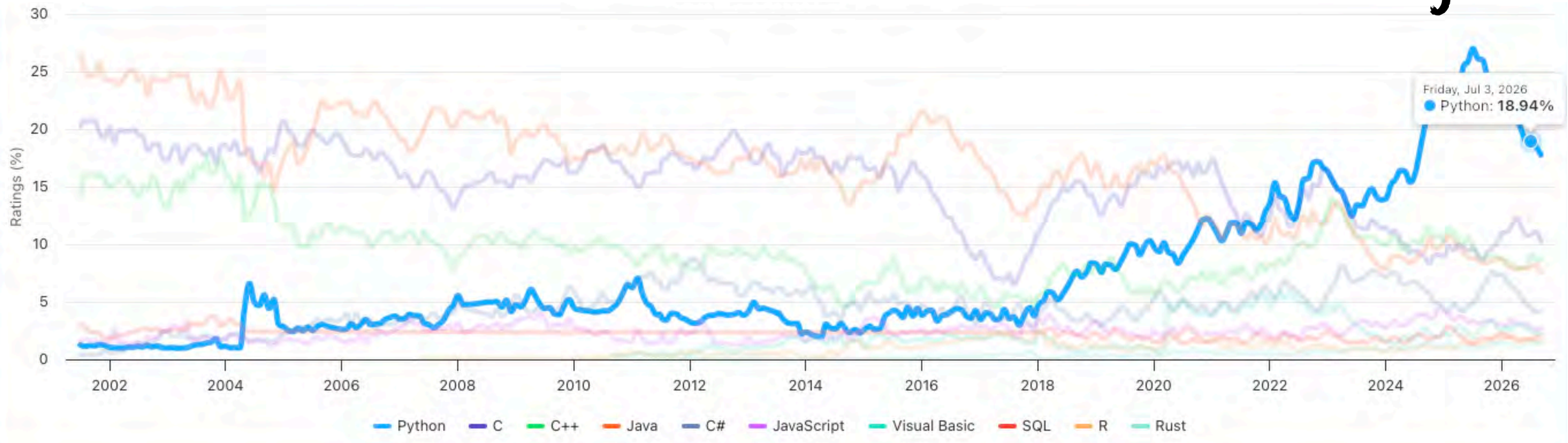


programming languages



TIOBE Programming Community Index Definition

https://www.tiobe.com/tiobe-index



The calculation consists in counting hits for the search query

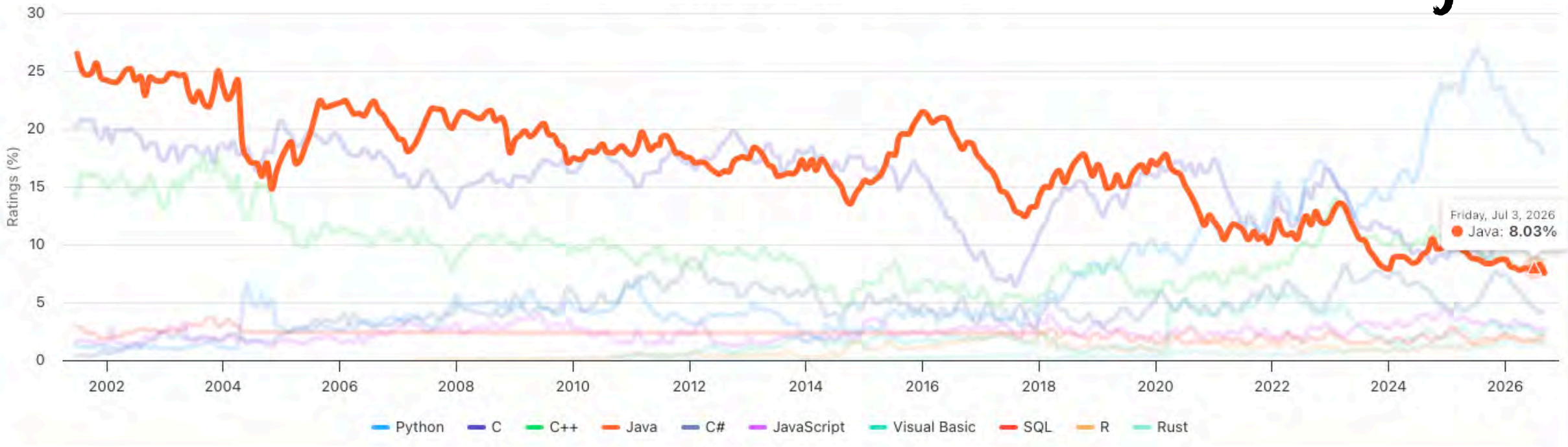


programming languages



TIOBE Programming Community Index Definition

https://www.tiobe.com/tiobe-index



The calculation consists in counting hits for the search query



python

programming languages



java

TIOBE Programming Community Index Definition

Programming Language	2026	2021	2016	2011	2006	2001	1996
Python	1	3	5	6	8	25	24
C	2	1	2	2	2	1	1
C++	3	4	3	3	3	2	2
Java	4	2	1	1	1	3	15
C#	5	5	4	5	7	13	-
JavaScript	6	7	7	10	9	9	20
Visual Basic	7	6	13	-	-	-	-
SQL	8	9	-	-	-	37	-
R	9	11	17	27	-	-	-
Delphi/Object Pascal	10	19	12	12	10	36	-
Ada	17	31	28	17	16	18	7
Lisp	32	33	27	13	13	17	8

The calculation consists in counting hits for the search query

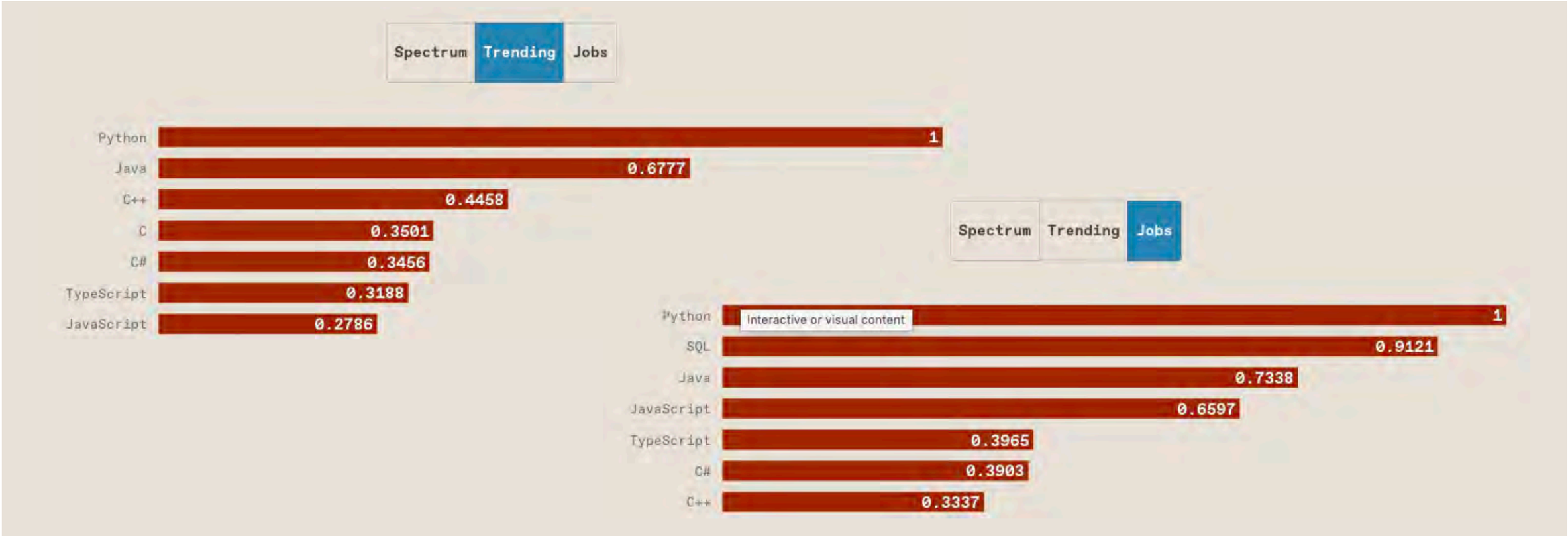


programming languages



The Top Programming Languages
IEEE Spectrum

https://spectrum.ieee.org/top-programming-languages-2025

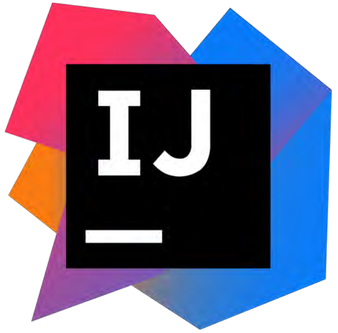


development tools



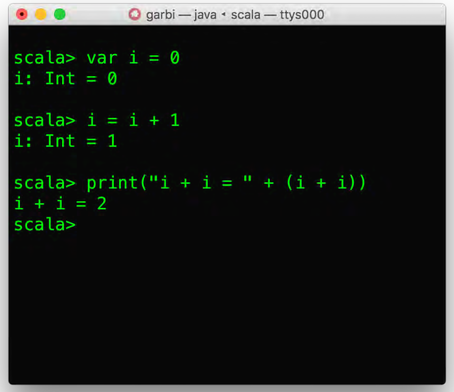
PyCharm

(python)



IntelliJ

(python / java)



the good old terminal